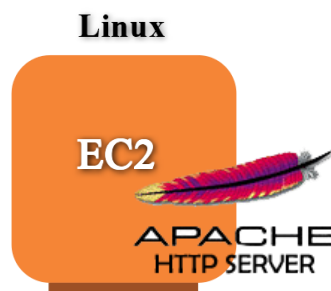




**AWS Solution Architect Training with
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**Trainer: Aravindraaj. G- Nminds Academy AWS
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**Configure Apache Web Server on Linux
Instance in AWS**



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Objective

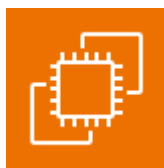
EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Linux instance is a virtual machine running the Linux operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, Apache, FileServer etc.) in the cloud without having to manage physical hardware.

Use Cases for EC2 Linux Instances:

- **Web and Application Hosting:** Run Apache web servers or other Linux-based web applications in the cloud.
- **Database Servers:** Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- **Remote Desktop Applications:** Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- **Development and Testing:** Create development environments with linux OS to test php, MySql or other Linux-based applications.

Used Services in AWS



Amazon EC2

Used Services in Linux Server

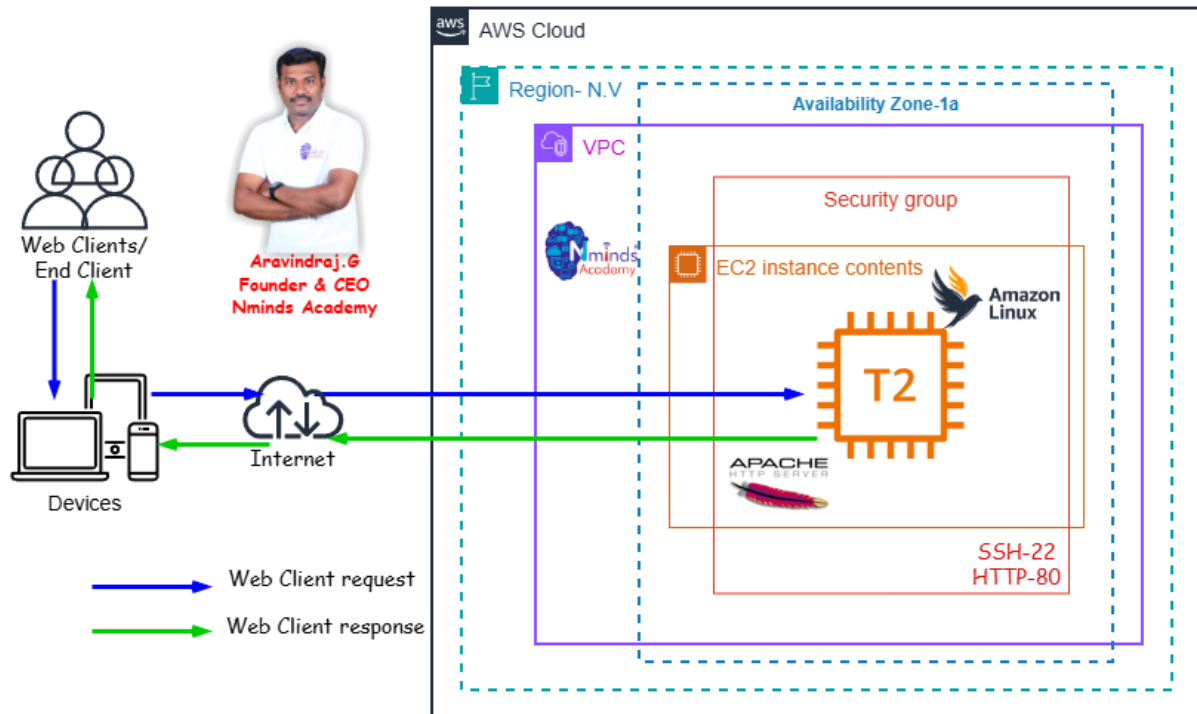


Apache Webserver



Topology

How to Configure the Apache web server in Linux Server with AWS EC2



The screenshot shows the AWS Management Console interface for the EC2 Instances page. The top navigation bar includes the AWS logo, a search bar, and various utility icons. The left sidebar contains a navigation menu with options like Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, and Capacity Manager. The main content area is titled 'Instances (1/1)' and includes a 'Launch instances' button. Below this, there's a 'Saved filter sets' section and a search bar. A table lists the instances, with the first instance 'i-02d90f3c4e1252822' in the 'Running' state. The instance details are expanded, showing the instance summary, public IPv4 address (98.83.153.84), private IPv4 address (172.31.38.130), and instance state (Running).

Instances (1/1) Info Last updated 1 minute ago

Connect Instance state Actions Launch instances

Saved filter sets Choose filter set Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status
☒	LINUX-K	i-02d90f3c4e1252822	Running	t3.micro	Initializing	View alarms +

i-02d90f3c4e1252822 (LINUX-K)

▼ Instance summary Info

Instance ID i-02d90f3c4e1252822	Public IPv4 address 98.83.153.84 open address	Private IPv4 addresses 172.31.38.130
IPv6 address 	Instance state Running	Public DNS

```
[root@ip-172-31-38-130:~]# ssh -o StrictHostKeyChecking=no ec2-user@98.83.153.84
```

```
-c cipher_spec [-D [bind_address:]port] [-E log_file]  
[-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]  
[-J destination] [-L address] [-l login_name] [-m mac_spec]  
[-O ctl_cmd] [-o option] [-P tag] [-p port] [-Q query_option]  
[-R address] [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]  
destination [command [argument ...]]
```

```
C:\Users\tauru>cd Downloads  
  
C:\Users\tauru\Downloads>ssh -i kec-group.pem ec2-user@98.83.153.84  
The authenticity of host '98.83.153.84 (98.83.153.84)' can't be established.  
ED25519 key fingerprint is SHA256:p0Q1BJg/FvgAUvAQfYCN14o12RiFYsQ06/bucRtMIhI.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '98.83.153.84' (ED25519) to the list of known hosts.
```

```
|_      #_  
|_|     #####          Amazon Linux 2023  
|_|    \#####\  
|_|     \####|  
|_|      \|/_-->  
|_|       V__'  
|_|      /___/  
|_|     _./_\_____  
|_|    _./_\_____  
|_|   _./_\_____  
|_|  _./_\_____  
|_| _./_\_____
```

```
[ec2-user@ip-172-31-38-130 ~]$ sudo -i  
[root@ip-172-31-38-130 ~]# yum update -y  
Amazon Linux 2023 repository                               69 MB/s | 61 MB   00:00  
Amazon Linux 2023 Kernel Livepatch repository            261 kB/s | 31 kB   00:00  
Dependencies resolved.  
Nothing to do.  
Complete!  
[root@ip-172-31-38-130 ~]#
```

```
root@ip-172-31-38-130:~  
Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13  
Installing : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13  
Installing : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 9/13  
Installing : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 10/13  
Installing : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 11/13  
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13  
Installing : httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13  
Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64 13/13  
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13  
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13  
Verifying : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64 3/13  
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13  
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13  
Verifying : httpd-2.4.66-1.amzn2023.0.1.x86_64 6/13  
Verifying : httpd-core-2.4.66-1.amzn2023.0.1.x86_64 7/13  
Verifying : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch 8/13  
Verifying : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64 9/13  
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13  
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13  
Verifying : mod_http2-2.0.27-1.amzn2023.0.3.x86_64 12/13  
Verifying : mod_lua-2.4.66-1.amzn2023.0.1.x86_64 13/13  
  
Installed:  
apr-1.7.5-1.amzn2023.0.4.x86_64  
apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64  
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  
httpd-core-2.4.66-1.amzn2023.0.1.x86_64  
httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch  
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64  
mailcap-2.1.49-3.amzn2023.0.3.noarch  
mod_lua-2.4.66-1.amzn2023.0.1.x86_64  
apr-util-1.6.3-1.amzn2023.0.2.x86_64  
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64  
httpd-2.4.66-1.amzn2023.0.1.x86_64  
httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch  
libbrotli-1.0.9-4.amzn2023.0.2.x86_64  
mod_http2-2.0.27-1.amzn2023.0.3.x86_64  
  
Complete!  
[root@ip-172-31-38-130 ~]#
```

WEB HOST OUTPUT

